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GitHub Link: <https://github.com/Felomondi/AI-and-future-of-work>

AI, Gender, and the Future of Work: Predicting Occupational Exposure to Artificial Intelligence

1. Introduction & Motivation

The rapid advancement of artificial intelligence is restructuring the labor market, but the resulting shifts are unlikely to be distributed equitably. Frey and Osborne (2017) estimated that 47% of U.S. employment was susceptible to computerization, but their framework focused on routine manual and cognitive tasks. The emergence of Large Language Models has shifted the frontier toward high-skill, white-collar occupations previously considered automation-proof.

Recent work makes the equity stakes concrete. The International Labour Organization (2025) finds women in high-income countries are nearly three times more likely than men to hold positions at the highest risk of AI-driven automation, attributed to overrepresentation in clerical and administrative roles. Felten, Raj, and Seamans (2021) introduced the AI Occupational Exposure (AIOE) index, which maps specific AI capabilities (translation, image recognition) to O*NET tasks. Muro et al. (2019) separately found that Hispanic workers were overrepresented in automation-prone occupations under the older Frey-Osborne framework.

Two gaps remain. First, the literature focuses on aggregate exposure rather than the specific job characteristics (cognitive vs. physical, text-heavy vs. spatial) that drive it. Second, applications of the AIOE index to racial demographics are sparse. This project uses machine learning to (1) identify which occupational traits predict AI exposure, and (2) determine whether the burden falls unevenly across demographic groups defined by gender and race.

Research Questions

- **RQ1:** Can we predict an occupation's level of AI automation exposure based on its task characteristics, skill requirements, and work context features?
- **RQ2:** Does the demographic composition of an occupation (gender and race) significantly predict AI exposure after controlling for job tasks and skills?

Expectations

We expect routine cognitive tasks (data entry, information processing) to predict higher AI exposure, while physical manipulation, social-emotional skills, and creative judgment predict lower exposure. For RQ2, we expect female-dominated occupations to show elevated language-modeling AI exposure (clerical and administrative concentration), Hispanic-dominated occupations to show lower exposure (physical labor and service concentration), Asian-dominated occupations to show modestly elevated exposure (STEM concentration), and Black-dominated occupations to fall between these poles. Each

direction carries equity implications: elevated exposure threatens existing wage gains, while lower exposure can reflect pre-existing structural barriers to higher-wage employment that AI may further entrench.

2. Data Description

Data Sources & Collection

We integrate four datasets, all linked at the 6-digit Standard Occupational Classification (SOC) code level:

- **O*NET Database (v30.2):** ~640 features across Skills, Abilities, Work Activities, and Work Context, on 0–7 or 1–5 importance/level scales. Primary predictor source.
- **Felten et al. AIOE Index (2021):** Three target variables (General AIOE, Language Modeling AIOE, Image Generation AIOE), constructed by mapping 10 AI applications to 52 O*NET abilities.
- **BLS Current Population Survey (2025):** Demographic shares per occupation (pct_women, pct_black, pct_hispanic, pct_asian).
- **BLS Occupational Employment and Wage Statistics (2024):** Median Annual Wages and Total Employment counts.

Data Preparation & Cleaning

- **SOC standardization:** O*NET's 8-digit codes were truncated to 6-digit codes for relational merging with BLS and AIOE data.
- **Reshaping:** O*NET files were pivoted from long to wide format before an inner join across all four sources.
- **Filtering:** A strict 20% missingness threshold was applied per O*NET feature, with target variables and demographic predictors protected from accidental drops.
- **Imputation & Scaling:** Remaining nulls in the 631 occupations were median-imputed; O*NET predictors were standardized (mean=0, variance=1) for Lasso/Ridge.

Variables for Modeling

The final dataset has 631 occupations and 650 variables. Targets: AIOE (primary), LMOE, IGOE. Predictors: 638 O*NET task features (RQ1), four BLS demographic shares (RQ2), median wage (RQ2 control).

3. Methods

We employ a multi-stage modeling framework, transitioning from unregularized linear baselines to regularized regression, tree-based ensembles, manifold learning, and finally mediation analysis with Double Machine Learning. RQ1 models use 5-fold cross-validation with randomized hyperparameter search.

3.1 Predictive Modeling and Regularization (RQ1)

An OLS baseline initially yielded $R^2 = 0.98$, which we diagnosed as data leakage from leaving LMOE/IGOE in the feature set. After isolating predictors, the corrected OLS achieved test $R^2 = 0.38$. The

high dimensionality and multicollinearity of the O*NET feature space justified our pivot to **Lasso (L1)** and **Ridge (L2)** regression. Lasso shrinks redundant coefficients to exactly zero, providing automated feature selection; Ridge stabilizes coefficient magnitudes when features are correlated. Lasso reduced the active feature set from 638 to 71 with a validation R^2 of 0.9755.

3.2 Non-Linear Ensembles (RQ1)

To capture interactions linear models miss, we fit **Random Forest** and **XGBoost** regressors. Random Forest reached validation $R^2 = 0.9613$; XGBoost reached 0.9720 with MAE = 0.116, building shallow trees (max_depth=3) and learning slowly (rate=0.05) to correct sequential errors.

3.3 Interpretability and Manifold Learning

SHAP (Shapley Additive Explanations) attributes each prediction to its constituent features in a mathematically consistent way. **UMAP** (Uniform Manifold Approximation and Projection) compresses the 638-dimensional task space into 2D, blind to the AIOE target, allowing us to verify whether occupations cluster by task similarity in ways that align with their AI exposure.

3.4 Demographic Analysis and Mediation (RQ2)

We test whether demographic composition predicts AIOE through six layered analyses, each addressing a limitation of the previous step:

1. **Pearson and Spearman correlations** of each demographic share with AIOE, LMOE, and IGOE, on both the full sample ($n=631$) and a matched-only subset ($n=193$, removing 438 occupations with imputed national-average demographics).
2. **Employment-weighted exposure** to estimate AIOE faced by the average worker in each group rather than the average occupation.
3. **Regression with wage controls** (OLS, Lasso, Ridge) to isolate demographic effects from the wage–AIOE correlation ($r = +0.49$).
4. **LMOE/IGOE disaggregation** to test whether different demographic groups face different *types* of AI exposure.
5. **Baron & Kenny (1986) mediation analysis** with seven mediators drawn from the top SHAP features in RQ1: four protective physical abilities (Extent Flexibility, Multilimb Coordination, Static Strength, Gross Body Coordination) and three cognitive risk factors (Written Comprehension, Written Expression, Memorization).
6. **Double Machine Learning (DML) robustness check** using the Partially Linear Regression framework (Chernozhukov et al., 2018). DML controls for the *full* 636-feature task space using XGBoost as the auxiliary learner, with 5-fold cross-fitting and 3 repetitions for stable inference.

Mediation gives an interpretable decomposition through hand-picked features; DML provides a stronger but less interpretable test using the full task space. Convergent results across both methods strengthen the joint conclusion.

4. Results

4.1 Primary Findings: The Drivers of AI Exposure (RQ1)

Model Performance

All four post-baseline models achieved test R^2 between 0.957 and 0.962, with **Lasso** (test $R^2 = 0.962$, MAE = 0.113) slightly leading and XGBoost close behind (test $R^2 = 0.960$, MAE = 0.114). Lasso shrunk the active feature set from 638 to 71. The convergence across linear and tree-based methods indicates that the underlying task→AIOE signal is strong and easily recovered once multicollinearity is controlled.

Interpretation via SHAP

SHAP analysis of XGBoost reveals a stark divide between cognitive and physical labor:

- **Risk factors (cognitive/textual tasks):** *Written Comprehension*, *Written Expression*, and *Memorization* are the primary drivers of automation vulnerability.
- **Protective factors (physical/manual tasks):** *Extent Flexibility*, *Multilimb Coordination*, *Gross Body Coordination*, and *Stamina* drive predicted exposure down.

The SHAP dependence plots reveal a sharp non-linear threshold: predicted AI exposure drops dramatically once an occupation requires moderate-to-high physical coordination, illustrating the current real-world limitation of AI in unstructured physical environments.

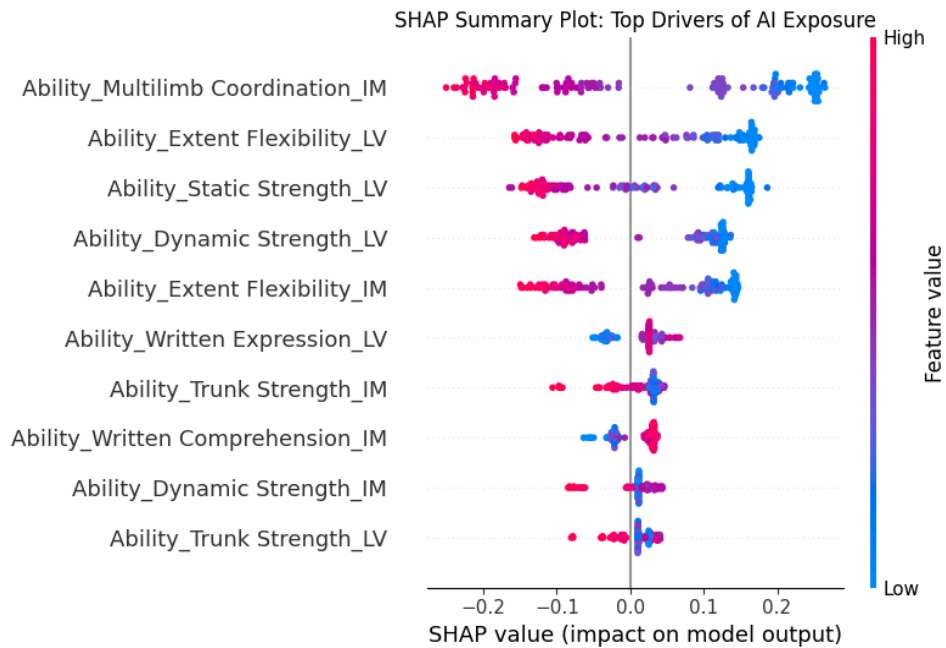


Figure 1: SHAP Summary Plot. Top features are physical abilities; high values (red) cluster on the negative SHAP side, confirming their protective effect.

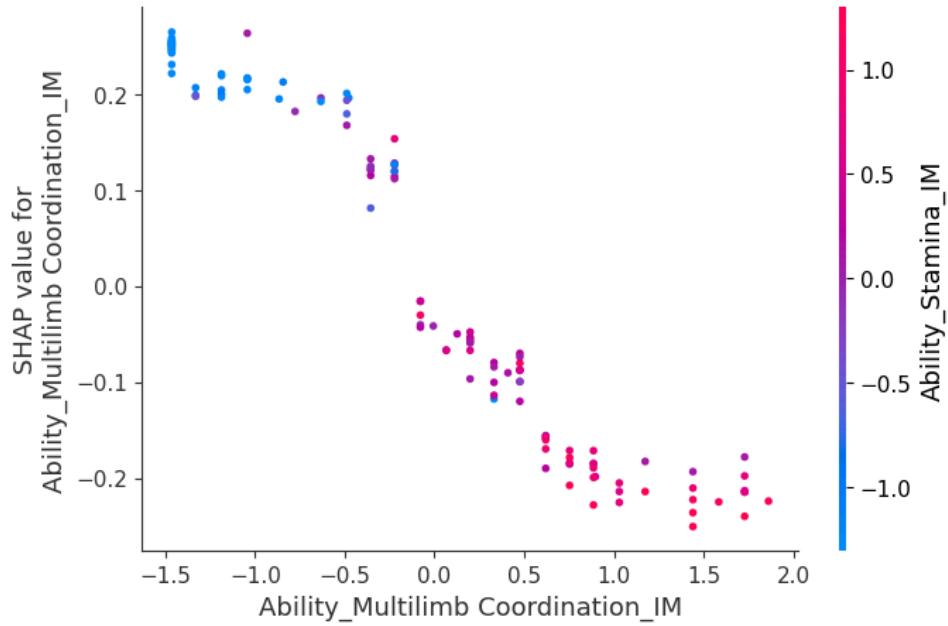


Figure 2: SHAP Dependence Plot for Multilimb Coordination, showing the sharp non-linear drop from +0.2 to -0.2 as the requirement crosses the population mean.

Visual Validation via UMAP

Projecting the 638-dimensional task space into 2D with UMAP and overlaying AIOE as a color map reveals a striking natural gradient: high-exposure (red) and low-exposure (blue) occupations separate into distinct clusters. Because UMAP was blind to the AIOE target, this clustering visually confirms that an occupation's task profile dictates its AI vulnerability.

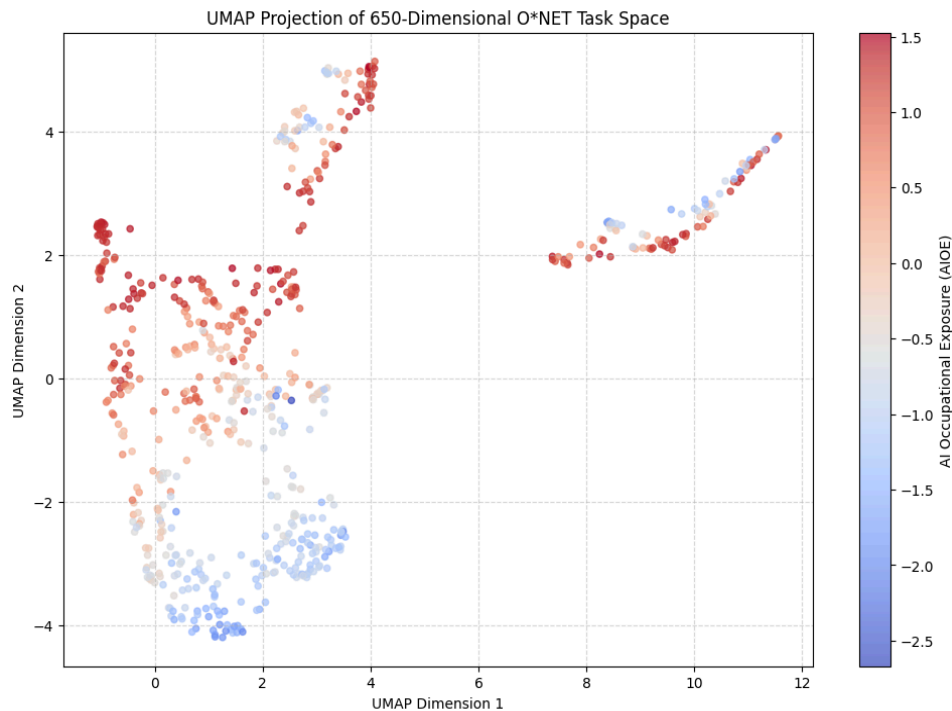


Figure 3: UMAP Projection of the O*NET Task Space, colored by AIOE. The red and blue regions emerge from task-feature similarity alone.

4.2 Demographic Disparities in AI Exposure (RQ2)

We test whether the burden of AI exposure falls unevenly across demographic groups in six layered analyses: bivariate correlation, employment-weighted exposure, regression with wage controls, LMOE/IGOE disaggregation, Baron-Kenny mediation, and a DML robustness check.

4.2.1 Bivariate Correlations

On the full sample (n=631), Pearson correlations are modest but mostly statistically significant; on the matched-only subset (n=193, where 438 occupations with imputed national-average demographics are removed), they strengthen considerably:

Demographic	AIOE (full, n=631)	AIOE (matched, n=193)	LMOE (full)	IGOE (full)
% Women	+0.152***	+0.316***	+0.181***	+0.027 n.s.
% Black	−0.094*	−0.214**	−0.067 n.s.	−0.175***
% Hispanic	−0.291***	−0.608***	−0.267***	−0.314***
% Asian	+0.140***	+0.215**	+0.121***	+0.157***
Median Wage	+0.494***	—	+0.445***	+0.556***

Table 1: Pearson correlations of demographic share with AIOE, LMOE, and IGOE. *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. = not significant. Matched subset excludes 438 occupations with imputed national-average demographics.

Three patterns stand out. **First**, the gender effect is asymmetric across AI types: women face elevated exposure to language-modeling AI (LMOE $r = +0.18$) but not image-generation AI (IGOE $r = +0.03$), consistent with their concentration in clerical, education, and legal-support roles. **Second**, Hispanic worker share shows the largest effect of any demographic variable, in the protective direction. The matched-sample correlation of -0.61 is larger in magnitude than the women–AIOE correlation of $+0.32$, making this the single strongest demographic–AIOE relationship in the data. **Third**, Black worker share shows a modest protective pattern (strongest for IGOE) while Asian worker share shows a modest elevated pattern across all AI types.

The matched-sample correlations are uniformly larger than the full-sample correlations, indicating that the full-sample numbers are *attenuated* by imputation noise rather than the matched estimates being inflated.

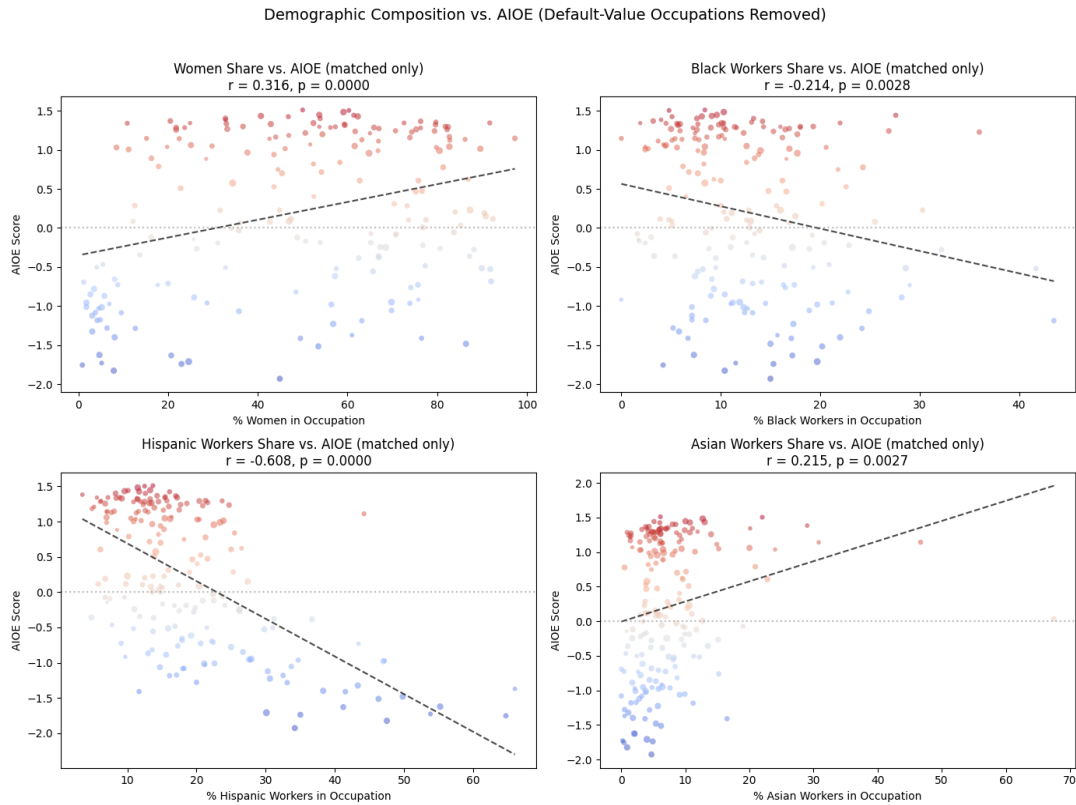


Figure 4: Demographic composition versus AIOE on the matched 193-occupation subset. The Hispanic–AIOE relationship (bottom-left) is the strongest of the four demographic groups.

4.2.2 Employment-Weighted Exposure

Weighting AIOE by employment share within each demographic group estimates the AIOE faced by the average worker rather than the average occupation.

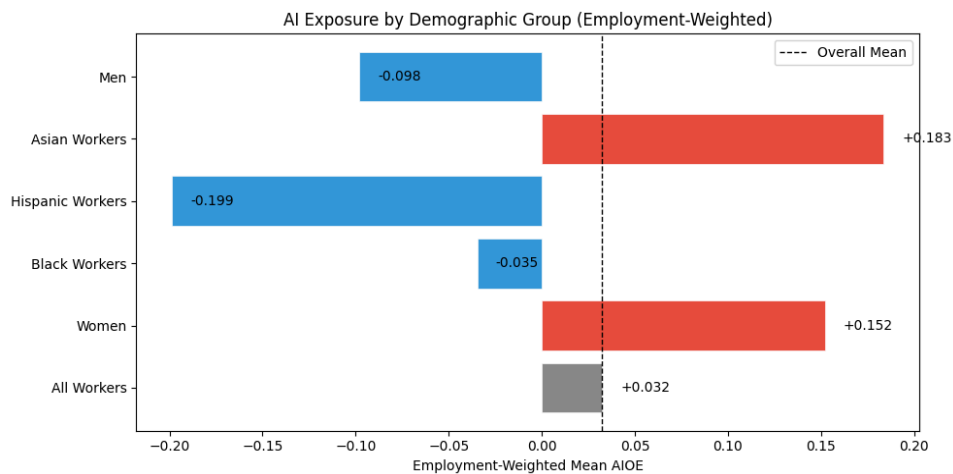


Figure 5: Employment-weighted mean AIOE by group. Asian workers face the highest exposure (+0.18 SD), Hispanic workers the lowest (−0.20 SD); a population-level gap of roughly 0.38 SD.

4.2.3 Regression with Wage Controls

Wages co-vary with both demographic composition and AIOE ($r=+0.49$ with median wage), so we fit OLS, Lasso, and Ridge regressions of standardized AIOE on the four demographic shares plus standardized median wage. All three estimators give nearly identical results. Standardized OLS coefficients:

- **% Women:** $\beta = +0.168$
- **% Hispanic:** $\beta = -0.122$
- **% Black:** $\beta = +0.010$ (near zero)
- **% Asian:** $\beta = +0.004$ (near zero)
- **Median Wage:** $\beta = +0.478$ (largest single effect)

After controlling for wage, the gender and Hispanic effects survive while the Black and Asian effects collapse to near zero. This indicates that the Black and Asian bivariate correlations are essentially wage-driven: Asian-dominated occupations correlate with AIOE because they tend to be high-wage cognitive-professional roles. The gender and Hispanic effects, in contrast, are independent of wage and warrant explanation through task structure.

4.2.4 LMOE vs. IGOE

Disaggregating AIOE into language-modeling and image-generation components reveals that demographic effects are not uniform across AI types.

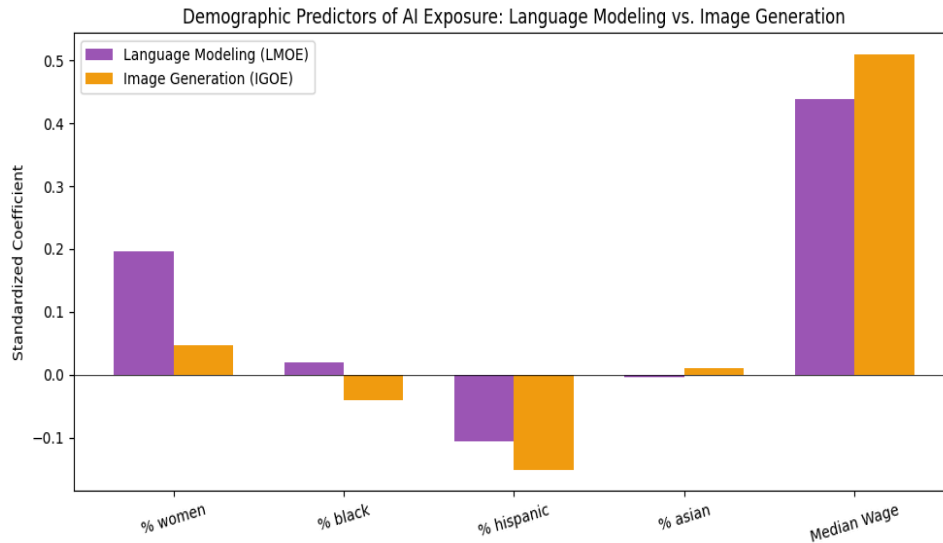


Figure 6: Standardized regression coefficients for LMOE vs. IGOE. The gender coefficient is roughly four times larger for LMOE (+0.20) than for IGOE (+0.05).

The most striking pattern is the **gender–AI type asymmetry**: women’s elevated exposure is concentrated where LLMs (writing assistants, summarization, customer service automation) have the most overlap, not where image generation is the relevant frontier. Hispanic worker share shows similarly negative coefficients for both AI types (−0.11 and −0.15), indicating a general protective effect. The Black coefficient flips sign across AI types (slightly positive for LMOE, slightly negative for IGOE).

4.2.5 Mediation Analysis

We test whether the surviving demographic effects are explained by *task structure* using Baron & Kenny (1986) mediation with seven mediators: four protective physical abilities and three cognitive risk factors drawn from the top SHAP features in RQ1.

Adding the seven task mediators raises R^2 from 0.30 (demographic + wage only) to 0.96, matching XGBoost’s RQ1 accuracy. The demographic coefficients change dramatically:

Predictor	Total β (c)	Direct β (c')	% Mediated
% Women	+0.168	−0.008	~105%
% Black	+0.010	+0.005	~53%
% Hispanic	−0.122	−0.032	~74%
% Asian	+0.004	−0.009	n/a (effect ~0)
Median Wage	+0.478	−0.018	~104%

Table 4: Baron & Kenny mediation analysis. Total effect is OLS β with wage controls; direct effect adds the 7 SHAP-derived task mediators.

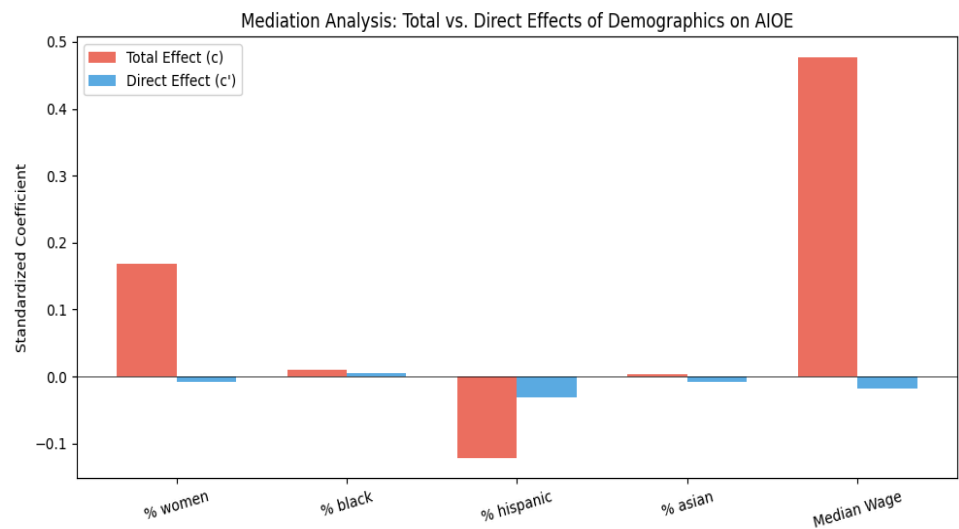


Figure 7: Mediation Analysis — Total vs. Direct Effects of Demographics on AIOE. The blue bars (direct effects with task mediators included) collapse to near zero across all demographic groups.

The Path-a regressions confirm the mechanism: women-dominated occupations require less physical labor ($\beta = -0.19$ to -0.21) and more text-processing skill ($\beta = +0.11$ to $+0.14$), while Hispanic-dominated occupations show the opposite pattern. **The demographic effects on AIOE are almost entirely mediated by task structure.** Women are not directly more exposed; their occupations cluster in the cognitive-textual task profile that maps onto current AI capabilities. The Hispanic protective effect runs in reverse: physical-labor concentration shields these occupations from the current AI frontier.

4.2.6 DML Robustness Check

A reasonable critic could argue that the seven mediators were cherry-picked from SHAP, leaving open omitted-variable bias. To rule this out, we estimate the residual demographic effects after controlling for the *full* 636-feature task space using Double Machine Learning with XGBoost as the auxiliary learner.

Demographic	OLS Total β	BK Direct β	DML 0 (full)	DML 95% CI
% Women	+0.168	−0.008	−0.002	[−0.017, +0.014]
% Black	+0.010	+0.005	−0.005	[−0.016, +0.007]
% Hispanic	−0.122	−0.032	−0.004	[−0.024, +0.017]
% Asian	+0.004	−0.009	+0.005	[−0.005, +0.015]

Table 5: Comparison of Total OLS, Baron-Kenny direct, and DML estimates. DML uses XGBoost auxiliary learners ($n_estimators=100$, $max_depth=3$), 5-fold cross-fitting, 3 repetitions. All four 95% CIs include zero in both the full sample and the matched subset ($n=193$).

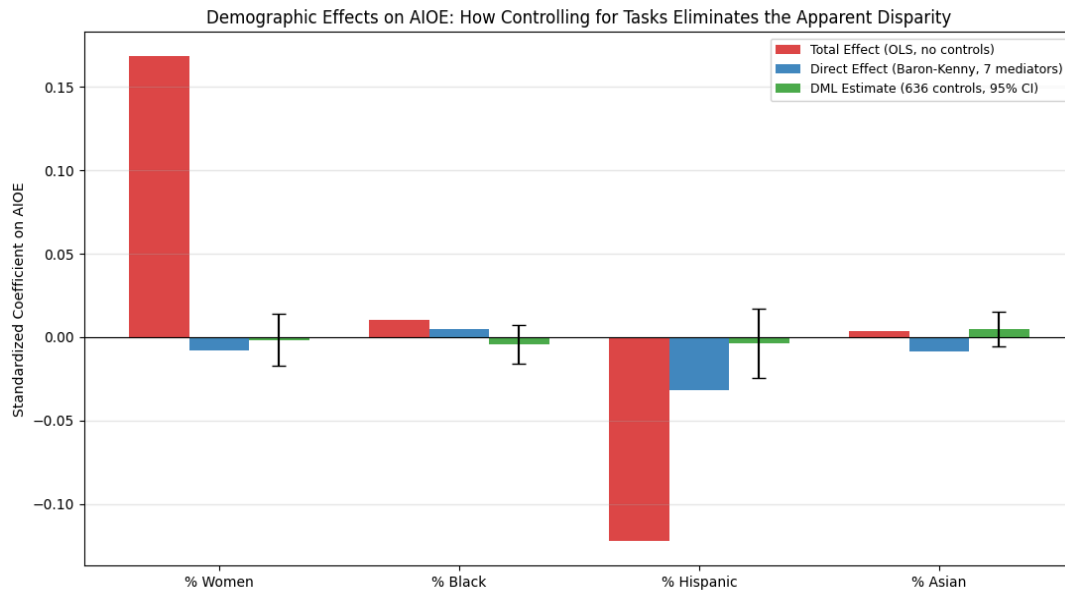


Figure 8: Demographic Effects on AIOE: Total (red) vs. Direct (blue, Baron-Kenny) vs. DML (green with 95% CI error bars). All DML confidence intervals cross zero.

The DML estimates are an order of magnitude smaller than the OLS total effects, and every 95% confidence interval includes zero. The Baron-Kenny finding survives the stronger test of controlling for the full 636-dimensional task space. Both methods converge: **once task structure is properly controlled, demographic composition has no residual predictive power for AIOE**. The DML on the matched subset ($n=193$) gives the same result, confirming the null is not driven by imputation noise.

4.3 Non-Results and Methodological Pivots

Two methodological failures shaped our pipeline. First, our initial OLS regression yielded a near-perfect R^2 of 0.98, which we diagnosed as data leakage from leaving LMOE and IGOE in the predictor set. Removing them collapsed performance to $R^2 = 0.38$, exposing the true challenge of the high-dimensional, multicollinear O*NET feature space and justifying our pivot to Lasso and Ridge. Second, for RQ2 we

initially considered raw correlation as the headline result. The substantial bivariate correlations (Hispanic $r = -0.61$) were later shown by mediation and DML to be downstream consequences of task segregation rather than direct demographic effects, prompting the layered six-stage analysis described in 4.2.

5. Discussion & Conclusion

5.1 Summary of Findings

First, an occupation's daily tasks are highly predictive of its AI Occupational Exposure. Lasso and XGBoost both achieved test R^2 of approximately 0.96, and SHAP analysis revealed a stark divide: cognitive and text-processing skills (Written Comprehension, Memorization) drive AI exposure upward, while physical attributes (Extent Flexibility, Multilimb Coordination, Static Strength) serve as protective shields. UMAP visually corroborated this divide.

Second, demographic composition correlates with AI exposure, but direction varies by group. Women face elevated exposure to language-modeling AI specifically (LMOE $\beta = +0.20$, four times the IGOE coefficient). Hispanic workers face the largest protective effect (matched-sample $r = -0.61$). Black workers face a modest protective pattern; Asian workers face a modest elevated pattern. At the worker level, the average Asian or female worker faces AIOE roughly 0.35 standard deviations above the average Hispanic worker.

Third, Baron-Kenny mediation and DML converge on a single conclusion: the demographic-AIOE relationships are almost entirely explained by occupational task characteristics. Baron-Kenny attributes 75-105% of the gender, Hispanic, and wage effects to seven task mediators; DML controls for all 636 task features and produces residual demographic effects whose 95% CIs all include zero. **Demographics do not directly cause AI exposure; occupational task segregation channels demographic groups into different task profiles, and AI capabilities map onto those profiles unevenly.**

5.2 Connection to Scholarly Literature

Our SHAP and UMAP analyses extend the Frey and Osborne (2017) framework to *generative AI*, confirming that the new automation frontier targets non-routine cognitive labor while physical labor remains insulated.

A notable contrast appears with Muro et al. (2019), who used Frey-Osborne computerization scores to argue that Hispanic workers were **over**represented in automation-prone occupations. Using the AIOE index, we reach the opposite conclusion: Hispanic-dominated occupations are the *most insulated* from current AI capabilities. This reflects a real shift in the technology frontier. The 2010s wave was driven by industrial robotics threatening routine manual labor; the 2020s generative AI wave targets cognitive-textual labor instead. The physical-labor concentration that made Hispanic workers vulnerable a decade ago now makes them protected.

Our results also provide structural backing to the ILO's (2025) warnings on the gendered impact of AI: women face higher exposure precisely because they are overrepresented in cognitive-routine and

text-processing tasks that LLMs replicate. The four-fold gap between gender–LMOE and gender–IGOE coefficients adds a finer point: the disparity is concentrated in language AI specifically.

5.3 Limitations

First, our analysis remains correlational. Even when controlling for the full task feature space, the underlying occupational segregation reflects deep historical patterns; demographic composition does not exist independently of the structural processes that determine which groups end up in which occupations. Second, *exposure* is not *displacement*: the AIOE index measures technological capability, not the economic, regulatory, or institutional frictions that determine actual job loss.

Data validity: Due to mismatches between O*NET and BLS taxonomies, 438 of 631 occupations received default national-average demographic values rather than unique matches. This attenuates correlation estimates: the Hispanic–AIOE correlation grows from -0.29 to -0.61 when we restrict to the 193 matched-only occupations. Importantly, our DML robustness check on the matched subset shows the null residual effect of demographics holds even where the bivariate signal is strongest, which strengthens the central mediation conclusion. Our analysis is also subject to the **ecological fallacy**: demographics aggregated at the occupational level cannot capture within-occupation variation (e.g., a female software engineer vs. a male software engineer in the same nominal role). Our models predict the AI exposure of standardized occupation profiles, not individual workers.

5.4 Ethical Concerns and Societal Implications

The finding that AI exposure is determined by task structure rather than demographic identity reframes the equity conversation: it is not that AI specifically targets any group, but that occupational segregation distributes the workforce into task profiles that AI capabilities map onto unevenly. This has direct policy implications.

Benefits: Because we have isolated the specific task features that act as risk factors (routine cognition, text processing) and protective shields (physical flexibility, complex coordination), workforce retraining can shift away from generic digital literacy toward targeted upskilling that complements AI. The mediation finding suggests interventions need to operate at the task-content level: changing what people *do* in their occupations is more leveraged than changing *who* fills them.

Risks: First, predictive AIOE scores could be misused as justification for preemptive layoffs in highly exposed, female-dominated sectors, treating theoretical exposure as inevitable displacement. Second, the apparent *protective* effect for Hispanic and Black workers carries an uncomfortable implication: low AI exposure correlates with concentration in low-wage physical-labor occupations, which itself reflects pre-existing structural barriers to professional employment. Lower AI exposure is therefore not unambiguously good news; it can reflect a labor market in which certain groups have been segregated into the work that machines are not yet capable of doing. Policy responses focused narrowly on high-exposure groups may inadvertently reinforce that segregation.

Future Work: Future analyses should resolve the ecological fallacy by acquiring micro-level individual-worker data, incorporate intersectional analyses (e.g., Hispanic women in clerical roles vs.

Hispanic men in construction), and use longitudinal data to determine whether exposure translates to actual displacement or augmentation.

5. What we could have done differently

In hindsight, several methodological choices either cost us time or limited the strength of our conclusions. We would address them as follows in a revised version of this project.

Resolve the BLS–O*NET taxonomy mismatch upfront. 438 of our 631 occupations received default national-average demographic values because BLS Current Population Survey categories do not map cleanly to 6-digit O*NET SOC codes. We discovered this only after running our first set of correlations and noticing vertical stripes in the scatter plots. The matched-only subset of 193 occupations gave us much stronger correlations (Hispanic r grew from -0.29 to -0.61), but the bulk of the dataset is essentially demographic noise. With more time, we would build a manual crosswalk from CPS occupation titles to SOC codes, or use the BLS’s OEWS file (which has finer SOC granularity) as a secondary demographic source.

Lead with mediation, not DML, for RQ2. Our first instinct was that DML was the "right" causal-inference tool because it sounded sophisticated, and our rough draft framed the entire RQ2 analysis around it. In practice, the Baron-Kenny mediation analysis (with seven hand-picked SHAP-derived task mediators) produced the more interpretable and substantively useful result: a clean decomposition showing 75-105% mediation. DML ended up being most valuable as a robustness check rather than the headline method. We would structure RQ2 around mediation as the primary tool and present DML as the confirmatory follow-up, which is the framing in this final report but was not the framing in our proposal.

Run DML with more repetitions from the start. Our first DML runs used a single repetition ($n_rep=1$) and produced point estimates that varied by ± 0.005 - 0.010 across runs due to stochastic cross-fitting. We initially mistook this for substantive instability before realizing it was just Monte Carlo noise. Setting $n_rep=3$ from the start would have saved a debugging cycle. With more compute, $n_rep=10$ would give even tighter standard errors.

Disaggregate RQ1 by AI type. We modeled AIOE as a single composite target for RQ1 and only disaggregated into LMOE and IGOE for the RQ2 demographic analysis. In hindsight, running SHAP on LMOE and IGOE separately would have been substantively interesting: do language-AI risk factors differ from image-AI risk factors? This is a natural extension that fits the existing pipeline and would strengthen the connection between RQ1 and RQ2.

Validate SHAP findings with permutation importance. SHAP and standard XGBoost feature importance gave consistent results in our analysis, but adding permutation importance as a third cross-check would make the RQ1 interpretive claims more defensible at minimal cost.

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